

The Burrau–Pythagorean Family

A discrete on-ramp to the full three-body IC manifold

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Abstract

We propose a structured exploration of the three-body problem’s initial-condition (IC) space via the *Burrau–Pythagorean family*: the infinite discrete set of right-triangle configurations whose side lengths form primitive Pythagorean triples, with masses assigned by the classical Burrau convention (each mass equals its opposite side length, normalised). This family is cleanly enumerable via Euclid’s parametrisation, maps directly into the Jacobi-coordinate IC pipeline of *Principia*, and provides a natural pedagogical escalation from a 0D discrete index through a 2D continuous mass simplex and 2D/4D momentum exploration (via invariant (L_z, E) charts or free Jacobi momenta) to the full 10D symmetry-reduced IC manifold. We outline the mathematical construction, the key hypothesis (that Pythagorean configurations may identify dynamically robust structures in the full IC space), and a “projective microscope” interaction mode for investigating high-dimensional basin structure around locked IC points.

1 Motivation

Burrau’s problem (1913) is a celebrated special case of the three-body problem: three bodies placed at the vertices of a 3–4–5 right triangle, with each mass equal to the length of its opposite side. It is one of the most studied chaotic trajectories in celestial mechanics, and its fractal basin boundaries are a canonical example of sensitive dependence on initial conditions.

However, the 3–4–5 triangle is merely the simplest member of an infinite family. Every primitive Pythagorean triple (a, b, c) defines a distinct right-triangle configuration, and the Burrau mass convention $m_i = \text{opposite side}$ (normalised so $\sum m_i = 1$) extends naturally to all of them. This note proposes using this family as a structured entry point into the full IC manifold explored by *Principia*, and outlines how relaxing the mass constraint opens a continuous landscape that connects to the unrestricted 10D exploration space.

2 The discrete Burrau–Pythagorean family

2.1 Euclid’s parametrisation of primitive triples

For coprime integers $m > n > 0$ with $m - n$ odd, the triple

$$a = m^2 - n^2, \quad b = 2mn, \quad c = m^2 + n^2 \tag{1}$$

is a primitive Pythagorean triple with hypotenuse c . Every primitive triple arises exactly once from such a pair (up to swapping legs $a \leftrightarrow b$). Non-primitive triples (ka, kb, kc) for $k > 1$

are related to the primitive case by scaling and are dynamically equivalent under the similarity symmetry of Newtonian gravity ($\mathbf{r} \rightarrow \lambda \mathbf{r}$, $t \rightarrow \lambda^{3/2} t$). We therefore restrict attention to primitive triples without loss of generality.

2.2 Distinct triangle shapes

Different primitive triples define genuinely different triangle shapes (i.e. non-similar triangles). The acute angles of a right triangle with legs a, b and hypotenuse c are $\arctan(a/b)$ and $\arctan(b/a)$. For example:

(m, n)	Triple (a, b, c)	Angles	Aspect ratio a/b
(2, 1)	(3, 4, 5)	36.9°, 53.1°	0.75
(3, 2)	(5, 12, 13)	22.6°, 67.4°	0.42
(4, 1)	(15, 8, 17)	61.9°, 28.1°	1.88
(4, 3)	(7, 24, 25)	16.3°, 73.7°	0.29

Each of these represents a qualitatively distinct initial configuration with different mass ratios and body separations.

2.3 Normalised Burrau configuration in Jacobi coordinates

Fix the right angle at the origin. Normalise the triangle by the hypotenuse so that it has unit size, and assign Burrau masses:

$$\mathbf{r}_1 = (0, 0), \quad m_1 = \frac{c}{a+b+c}, \quad \mathbf{r}_2 = \left(\frac{a}{c}, 0\right), \quad m_2 = \frac{b}{a+b+c}, \quad \mathbf{r}_3 = \left(0, \frac{b}{c}\right), \quad m_3 = \frac{a}{a+b+c}. \quad (2)$$

The Jacobi vectors follow directly. Taking bodies 1–2 as the inner pair:

$$\boldsymbol{\rho} = \mathbf{r}_2 - \mathbf{r}_1 = \left(\frac{a}{c}, 0\right), \quad (3)$$

$$\boldsymbol{\lambda} = \mathbf{r}_3 - \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2} = \left(-\frac{ab}{c(b+c)}, \frac{b}{c}\right). \quad (4)$$

These expressions, together with the normalised masses, define a unique point in the symmetry-reduced IC manifold of Principia for each primitive (m, n) pair.

2.4 The family as a discrete subset of the IC manifold

Since the masses are locked to the geometry by the Burrau convention, each primitive triple (m, n) specifies *all* of the configuration and mass degrees of freedom simultaneously. In the language of the Principia IC pipeline, this family is a countable discrete subset of the full 10D latent chart $\mathbf{z} \in \mathbb{R}^{10}$, with all momentum coordinates set to zero (the classical Burrau initial condition is a rest start). The shape of each configuration is fully characterised by the ratio $|\boldsymbol{\lambda}|/|\boldsymbol{\rho}|$ and the angle between the Jacobi vectors, both of which vary across primitive triples.

3 Relaxing the mass constraint: the continuous extension

3.1 The mass simplex at fixed geometry

The strict Burrau convention locks the mass ratios to the triangle geometry. But there is no physical reason to maintain this coupling. For a given primitive triangle shape, we can decouple the masses and allow them to vary freely over the 2-simplex

$$\Delta_2 = \{(m_1, m_2, m_3) \in \mathbb{R}_{>0}^3 : m_1 + m_2 + m_3 = 1\}. \quad (5)$$

The Burrau point sits at one distinguished location in this simplex:

$$(m_1^B, m_2^B, m_3^B) = \frac{1}{a+b+c}(c, b, a). \quad (6)$$

For each primitive triple, the mass simplex provides a 2D continuous family of initial conditions sharing the same triangle geometry but with different mass assignments.

3.2 Momentum as an independent exploration axis

The classical Burrau problem starts from rest ($\mathbf{p}_i = \mathbf{0}$), but there is no physical reason to restrict to zero momentum. Because the Principia pipeline factors the IC decode into independent mass, configuration, and momentum stages, momentum exploration composes freely with the mass and geometry variations described above.

Free Jacobi momenta. The most general option: decode two planar Jacobi momentum vectors $(\mathbf{p}_\rho, \mathbf{p}_\lambda) \in \mathbb{R}^4$ from four free controls, as in the main Principia IC document. The zero-momentum rest start sits at the origin of this 4D space. For a fixed triangle and fixed masses, this gives a 4D family; a 2D rendering slice through it shows how the basin structure changes as the system is given an initial kick.

Invariant momentum charts: (L_z, E) and (L_z, K) . These are particularly natural for the Burrau family. The classical rest start corresponds to $(L_z, E) = (0, U)$ —the absolute bottom of the feasibility parabola, where all energy is potential and there is no angular momentum. The invariant chart then fans upward and outward from this single point as the energy budget grows:

$$|L_z| \leq \sqrt{2I(E - U)}, \quad (7)$$

so the rest start is the apex of the feasible region viewed from below. A 2D (L_z, E) rendering at a fixed Burrau configuration would show how the fractal basins nucleate at rest and evolve as the system acquires angular momentum and kinetic energy—a visualisation of how the classical Burrau problem connects to more energetic dynamics.

The (L_z, K) chart is equivalent but parameterises kinetic energy directly, which is sometimes more convenient: the rest start sits at $(L_z, K) = (0, 0)$, the corner of the feasibility wedge.

Geometry–momentum views. Another natural 2D slice pairs a geometric coordinate (e.g. the relative Jacobi angle β , or the shape parameter α) against a momentum coordinate (e.g. L_z or total kinetic energy K). This shows how the basin structure changes jointly in shape and energy—interpolating between a “pure Burrau” axis and a “pure kick” axis.

3.3 Paper structure: progressive revelation

The full decomposition of degrees of freedom suggests a natural escalation:

Stage 1 (discrete, 0D per triple). The strict Burrau–Pythagorean family at rest. For each primitive (m, n) , render the fractal basin boundary at zero initial momentum. Classify how basin topology (ejection prevalence, collision structure, bounded-orbit regions) varies across primitive triples. This is already novel: no systematic survey of Burrau-type dynamics across Pythagorean triples exists in the literature.

Stage 2a (continuous masses, 2D per triple). Fix a triangle shape and sweep the mass simplex Δ_2 at zero momentum. The Burrau point becomes a distinguished landmark in a 2D landscape. Key questions: How do the basins deform as masses are dragged away from the Burrau assignment? Is the Burrau point dynamically special (e.g. a critical point of some chaos measure, a local extremum of ejection prevalence)?

Stage 2b (continuous momentum, 2D per triple). Fix a triangle shape and Burrau masses, and sweep the (L_z, E) or (L_z, K) invariant chart. The rest start sits at the apex/corner of the feasibility region; the rendering reveals how basins nucleate and grow as energy and angular momentum increase. Alternatively, use the full 4D free-momentum space and take 2D slices.

Stage 3 (combined, 4D+ per triple). Allow masses and momenta to vary simultaneously. The Burrau rest-start point is now a single distinguished landmark in a space of dimension up to $2 + 4 = 6$ (mass simplex $\times$ momentum). Two-dimensional slices through this joint space reveal correlations between mass assignment and momentum structure that are invisible when each is varied in isolation.

Stage 4 (full, 10D). Remove all constraints and explore the complete symmetry-reduced IC manifold. The Burrau–Pythagorean family and its mass/momentum extensions become particular cross-sections of this manifold, providing interpretable reference points within the full high-dimensional space.

4 The projective microscope: rotating slices through a locked IC

4.1 Concept

Fix a point $\mathbf{z}_0 \in \mathbb{R}^{10}$ in the IC manifold (the “locked pixel”). The Principia renderer displays 2D slices defined by affine chart maps

$$\mathbf{z}(s, t) = \mathbf{z}_0 + (2s - 1) \mathbf{q}_1 + (2t - 1) \mathbf{q}_2, \quad (s, t) \in [0, 1]^2. \quad (8)$$

The *projective microscope* interaction allows the user to keep $\mathbf{z}_0$ fixed while changing the basis vectors $(\mathbf{q}_1, \mathbf{q}_2)$ that define the slicing plane. This is equivalent to rotating a 2D viewing plane through a fixed point in the 10D manifold.

4.2 Axis-aligned slices

The simplest version restricts $\mathbf{q}_1$ and $\mathbf{q}_2$ to be standard basis vectors $\mathbf{e}_i, \mathbf{e}_j$ (scaled by appropriate half-ranges). This gives $\binom{10}{2} = 45$ distinct axis-aligned views through any given point, each showing how the basin structure varies as two specific physical parameters (e.g. a mass ratio

and a momentum component, or two configuration angles) change while all others are held fixed.

4.3 Smooth interpolation between slicing planes

More generally, one can smoothly interpolate between any two slicing orientations. Let $(\mathbf{q}_1^{(A)}, \mathbf{q}_2^{(A)})$ and $(\mathbf{q}_1^{(B)}, \mathbf{q}_2^{(B)})$ define two 2D planes through $\mathbf{z}_0$. A smooth interpolation can be constructed (e.g. via geodesic interpolation on the Grassmannian $\text{Gr}(2, 10)$, or more practically by interpolating and re-orthonormalising the basis vectors). As the user morphs between planes, the fractal basin structure around the locked point continuously deforms, revealing how structures visible in one cross-section connect to structures in another.

5 Burrau-specific charts

The following charts are specialised to the Burrau–Pythagorean setting but are implemented as standard chart maps $\Phi : [0, 1]^2 \rightarrow Y$ feeding into the existing decoder and canonicaliser pipeline. No modifications to the renderer, integrator, or event-detection logic are required.

5.1 The Euclid plane

Use the Euclid parameters (m, n) as continuous axes. Given $(s, t) \in [0, 1]^2$, define

$$m(s) = 2 + (m_{\max} - 2)s, \quad n(t) = 1 + (m(s) - 1 - \varepsilon)t, \quad (9)$$

where the bounds ensure $m > n > 0$ everywhere and $m_{\max}$ sets the range of triples explored. The Euclid formulae (1) then produce a continuous triangle shape at every pixel, and the chart map feeds into the standard Jacobi configuration decoder via the normalised positions (2).

At integer lattice points satisfying $\text{gcd}(m, n) = 1$ and $m - n$ odd, the pixel lands exactly on a primitive Pythagorean triple. At non-integer points, it produces a non-Pythagorean right triangle—a perfectly valid three-body configuration, but one that is not number-theoretically distinguished. The primitive triples therefore appear as highlighted landmarks on a continuous landscape.

This chart directly tests whether the Pythagorean configurations sit at dynamically special locations (e.g. local extrema of ejection prevalence, or bifurcation points in the basin topology) or whether they blend smoothly into the background of generic right triangles.

Mass assignment. At each point in the Euclid plane, the Burrau convention assigns $m_i =$ opposite side (normalised). Since the side lengths vary continuously with (m, n) , the masses also vary continuously across the chart. Alternatively, a fixed mass assignment can be used to isolate the effect of geometry from the effect of mass ratios.

5.2 The acute-angle axis

Every right triangle is determined up to similarity by its smaller acute angle $\theta \in (0, \pi/4]$ (restricting to $\theta \leq \pi/4$ removes the leg-swap redundancy). This gives a clean 1D parameterisation of all

right-triangle shapes, not just Pythagorean ones. Given θ , the normalised legs are $(\cos \theta, \sin \theta)$ and the hypotenuse is 1.

In a 2D chart, place θ on one axis and a second physical parameter on the other:

$$\theta(s) = \varepsilon_\theta + \left(\frac{\pi}{4} - 2\varepsilon_\theta\right)s, \quad s \in [0, 1], \quad (10)$$

with a small buffer ε_θ to avoid the degenerate limits $\theta \rightarrow 0$ (infinitely elongated) and $\theta \rightarrow \pi/4$ (isosceles right triangle). Natural choices for the second axis include:

- L_z or K : shows how the basin structure varies jointly in triangle shape and energy/angular momentum.
- A mass perturbation coordinate: deviation from the Burrau mass assignment along a geodesic in the mass simplex.
- The integration horizon T : reveals progressive refinement of the bounded/escape classification as simulation time increases.

The primitive Pythagorean triples appear as distinguished vertical lines at their specific θ values (e.g. $\theta_{3,4,5} = \arctan(3/4) \approx 36.9^\circ$, $\theta_{5,12,13} = \arctan(5/12) \approx 22.6^\circ$). The rendering immediately shows whether the basin structure varies smoothly with θ or has bifurcations at or near the Pythagorean angles.

5.3 The ternary mass plot

For a fixed triangle geometry, the mass simplex Δ_2 is itself a 2D domain. We parameterise it via barycentric coordinates

$$(m_1, m_2, m_3) = (1 - s - t, s, t), \quad s, t \geq 0, \quad s + t \leq 1, \quad (11)$$

or equivalently via the two-logit softmax decode already defined in the Principia pipeline ($z_{\mu 1}, z_{\mu 2}$ as exploration axes with all other latent coordinates fixed).

The Burrau point $(c, b, a)/(a + b + c)$ is a single marked location in this simplex. The rendering shows how the basin classification—or a scalar chaos measure such as the frequency diffusion D —varies as masses move away from the Burrau assignment at fixed geometry and (typically) zero momentum.

This chart is particularly effective when animated across primitive triples: each frame is a different triangle geometry, same mass simplex, and the viewer watches how the Burrau point’s local basin environment changes with the underlying shape.

5.4 Bifurcation strips

This is a compact chart designed for publication-quality figures. One axis indexes triangle shapes (ordered by θ , or by hypotenuse c , or by Euclid parameter m); the other axis is a single continuous parameter such as K , L_z , or a mass perturbation scalar.

Concretely, let the horizontal axis $s \in [0, 1]$ map to the acute angle $\theta(s)$ as in §5.2, and let the vertical axis $t \in [0, 1]$ map to kinetic energy $K(t) = K_{\max} t^{\gamma_K}$. Each vertical column is a 1D slice of the basin structure for one triangle shape at increasing energy. The resulting image

is a bifurcation-diagram-style visualisation: the horizontal axis sweeps triangle shape, and the vertical axis shows onset of chaos, ejection thresholds, or prevalence of bounded orbits for each shape.

This chart uses the standard Principia pipeline unchanged. The only specialisation is the chart map Φ , which constructs the right-triangle Jacobi configuration from $\theta(s)$ and feeds $K(t)$ into the $(L_z = 0, K)$ invariant momentum decode (or, if nonzero angular momentum is desired, a fixed L_z can be set as a parameter).

Discrete vs. continuous horizontal axis. The horizontal axis can be continuous (every pixel a different right-triangle shape, as above), or discrete (each column a specific primitive triple, with the column width representing one “pixel” per triple). The discrete version is useful for direct comparison across the family; the continuous version reveals whether the discrete triples sit at dynamically distinguished locations.

6 Central hypothesis: Burrau configurations as probes of high-dimensional structure

6.1 Statement

Hypothesis. The fractal basin structures observed in the Burrau–Pythagorean family are cross-sections of codimension- k structures ($k < 8$) in the full 10D IC manifold, rather than artefacts of the constrained submanifold. In particular, basin boundaries that appear as 1D curves in the 2D Burrau slice persist as higher-dimensional surfaces when the slicing plane is rotated into unconstrained directions.

6.2 How to test it

The projective microscope provides a direct test. Lock onto a basin boundary identified in a Burrau slice (a point $\mathbf{z}_0$ on or near a boundary between, say, escape-body-1 and escape-body-2 basins). Then:

1. Begin with the slice basis vectors spanning the Burrau-constrained submanifold.
2. Continuously rotate the slice basis to include directions that point into unconstrained dimensions (e.g. independent mass variation, or momentum perturbation).
3. Observe whether the basin boundary persists, deforms smoothly, or dissolves.

If the boundary persists and deforms smoothly, it is a genuine feature of the full IC space that the Pythagorean cross-section happened to reveal. If it dissolves, the structure is specific to the constrained submanifold.

6.3 Significance

A positive result (persistence) would establish that number-theoretic initial conditions—specifically, the Pythagorean triples—are identifying dynamically meaningful structures in the full IC space. This would be a connection between number theory and Hamiltonian dynamics that, to our

knowledge, has not been previously demonstrated. A negative result (dissolution) is also informative: it would characterise the Burrau family as a dynamically special (measure-zero) subset with qualitatively different basin topology from generic nearby configurations.

7 Implementation notes

7.1 Compatibility with the Principia pipeline

The constructions described here are fully compatible with the existing Principia IC pipeline. Specifically:

- **Mass decode.** The Burrau masses $(c, b, a)/(a + b + c)$ are simply specific values of the softmax mass logits. The continuous mass extension corresponds to sweeping the mass logits while holding configuration controls fixed.
- **Configuration decode.** The Jacobi vectors (3)–(4) map to specific values of the hyperspherical Jacobi parameters $(\alpha, \beta, \tilde{R})$. The triangle selector (primitive (m, n) pair) determines these values; no new decode logic is required.
- **Slice basis rotation.** The projective microscope is a UI layer over the existing affine slicing mechanism: $\mathbf{z}_0$ is pinned and $(\mathbf{q}_1, \mathbf{q}_2)$ are updated. The GPU integration kernel is unchanged.
- **Momentum.** The classical Burrau problem uses zero initial momenta, but the pipeline naturally supports nonzero momenta via free Jacobi momentum decode or invariant $(L_z, E)/(L_z, K)$ charts. The rest start $(L_z, E) = (0, U)$ is a single distinguished point in the invariant chart’s feasibility region; sweeping away from it requires no new construction beyond what is already defined in the main IC document. The momentum decode is independent of the mass and configuration decodes, so all combinations compose freely.

7.2 Enumeration bounds

The number of primitive Pythagorean triples with hypotenuse $c \leq N$ grows as $\mathcal{O}(N/\log N)$. For practical purposes, iterating over (m, n) pairs with $m \leq 50$ yields approximately 400 primitive triples, which is more than sufficient for a systematic survey. The computational cost per triple is one full basin-boundary render (identical to the cost of any single Principia frame).

8 Summary

The Burrau–Pythagorean family provides:

1. A **discrete, number-theoretically indexed** set of three-body configurations that are cleanly enumerable and map directly to the Principia IC pipeline.
2. A natural **progressive escalation** from constrained (0D) through partially relaxed (2D mass simplex, 2D momentum invariant charts, or 4D free momentum) to the full unconstrained (10D) IC manifold. Each stage adds independent degrees of freedom, with the classical Burrau rest start as a distinguished landmark at every level.

3. A concrete, testable **hypothesis** about whether Pythagorean configurations identify robust dynamical structures in the full IC space.
4. A novel interaction mode (the **projective microscope**) for investigating high-dimensional basin topology by rotating 2D slices through fixed points.

Together, these form a self-contained research programme that is implementable with the existing Principia architecture and that connects classical celestial mechanics, number theory, and high-dimensional dynamical systems in a way that—to our knowledge—has not been previously explored.

A Projective microscope: implementation

This appendix describes the projective microscope interaction mode introduced in §4. The existing Principia UI already provides per-dimension sliders for the 10 latent coordinates (controlling $\mathbf{z}_0$) and a tilt mechanism that continuously rotates the 2D slice plane into chosen dimensions. The microscope adds one new primitive on top of this: **locking the slice centre to a specific IC**, so that all existing controls—sliders, tilt, chart mode, zoom—orbit around a fixed point instead of floating freely.

A.1 The lock gesture

When the user selects a pixel at texture coordinates $(s, t) \in [0, 1]^2$, we snap the slice centre to the IC at that pixel.

Affine charts. Compute the locked coordinate CPU-side:

$$\mathbf{z}_{\text{locked}} = \mathbf{z}_0 + (2s - 1) \mathbf{q}_1 + (2t - 1) \mathbf{q}_2. \quad (12)$$

Write $\mathbf{z}_{\text{locked}}$ into all 10 slider values simultaneously. No GPU readback is required—the current $(\mathbf{z}_0, \mathbf{q}_1, \mathbf{q}_2)$ are already known on the CPU.

Nonlinear charts. For charts involving nonlinear domain warps (e.g. the (L_z, E) feasibility warp, or the Euclid plane), the mapping from (s, t) to the 10D coordinate is nonlinear. Two options: (i) replicate the chart map and decoder on the CPU and evaluate at the selected (s, t) ; or (ii) write the full decoded $\mathbf{z}$ per pixel to a storage buffer during the compute pass and read back the selected entry. Option (i) is preferred when the decode logic is not excessively complex.

A.2 What changes in locked mode

Very little. The GPU shader is completely unchanged—it reads $(\mathbf{z}_0, \mathbf{q}_1, \mathbf{q}_2)$ from the uniform buffer identically in both modes. The only difference is in how the CPU-side input handler routes user actions:

Sliders. In free mode, each slider directly sets one component of $\mathbf{z}_0$. In locked mode, the sliders are frozen (or hidden): $\mathbf{z}_0 = \mathbf{z}_{\text{locked}}$ is pinned.

Tilt. Works identically in both modes. The tilt mechanism already rotates the slice plane into chosen dimensions by a controllable amount—in locked mode, it simply does so around the locked point rather than the current slider position. This is the core of the “different perspectives on the same IC” interaction.

Chart mode. Switching chart mode (e.g. from the latent slice to the (L_z, E) invariant chart, or to the Euclid plane) changes the chart map Φ but keeps $\mathbf{z}_{\text{locked}}$ as the centre. The locked IC remains at the centre pixel; what changes is which degrees of freedom the surrounding pixels explore. This lets the user ask: “what does the neighbourhood of this IC look like when I vary energy vs. when I vary geometry?”

Zoom. Works identically in both modes (scales $\mathbf{q}_1$ and $\mathbf{q}_2$ by a common factor).

Unlock. On unlock, the current view state (centre, basis, zoom) is preserved as the new free-mode starting point. The sliders resume normal operation from the locked position.

A.3 Named compound directions

The one genuinely new UI element the microscope motivates is a library of **named slice directions**: pre-defined $\mathbf{q}$ vectors that correspond to physically meaningful explorations around a locked point. These compose with the existing tilt mechanism (which interpolates between the current basis and a target direction).

Mass perturbation away from Burrau. Given the Burrau mass logits (μ_1^B, μ_2^B) , define a direction in the mass subspace pointing toward equal masses:

$$\mathbf{d}_{\text{mass}} = (0, 0, 0, 0, -\mu_1^B, 0, 0, 0, 0, -\mu_2^B) \in \mathbb{R}^{10}, \quad (13)$$

normalised to unit length. Tilting the slice into $\hat{\mathbf{d}}_{\text{mass}}$ sweeps masses away from the Burrau assignment while keeping the locked geometry and momentum fixed at the centre pixel.

Energy increase at fixed angular momentum. A direction in the momentum subspace that increases kinetic energy without changing L_z . This is configuration-dependent (it depends on the decoded positions at the locked point) and must be recomputed when the locked IC changes. The construction follows the projected direction field from the (L_z, E) momentum decode in the main IC document.

Burrau-to-unconstrained morph. Start with the tilt basis spanning the Burrau-constrained submanifold (e.g. configuration angles α and β at fixed Burrau masses and zero momentum). The named direction library includes alternatives that point into mass or momentum subspaces. Using the existing tilt slider to interpolate from the constrained basis toward an unconstrained direction directly implements the hypothesis test from §6: the user watches the basin boundary persist, deform, or dissolve as the slice plane rotates out of the constrained submanifold.

A.4 Summary of new implementation work

Component	Status	New work
Affine slice uniforms	Existing	None
Per-dimension sliders	Existing	Freeze/unfreeze in locked mode
Tilt mechanism	Existing	None (works unchanged)
Chart mode switching	Existing	None (works unchanged)
Zoom	Existing	None
Lock gesture	New	Pixel $\rightarrow$ $\mathbf{z}$ readback, pin $\mathbf{z}_0$
Named direction library	New	Precomputed $\mathbf{q}$ vectors, menu UI
GPU shader	Existing	No changes